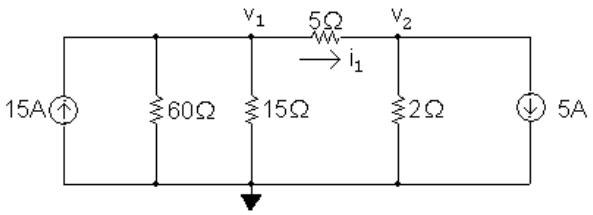
4

Techniques of Circuit Analysis

Assessment Problems

AP 4.1 [**a**] Redraw the circuit, labeling the reference node and the two node voltages:



The two node voltage equations are

$$-15 + \frac{v_1}{60} + \frac{v_1}{15} + \frac{v_1 - v_2}{5} = 0$$

$$5 + \frac{v_2}{2} + \frac{v_2 - v_1}{5} = 0$$

Place these equations in standard form:

$$v_1\left(\frac{1}{60} + \frac{1}{15} + \frac{1}{5}\right) + v_2\left(-\frac{1}{5}\right) = 15$$

$$v_1\left(-\frac{1}{5}\right) + v_2\left(\frac{1}{2} + \frac{1}{5}\right) = -5$$

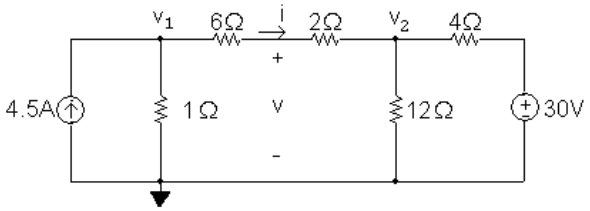
Solving, $v_1 = 60$ V and $v_2 = 10$ V;
Therefore, $i_1 = (v_1 - v_2)/5 = 10$ A

[**b**] $p_{15\text{A}} = -(15\,\text{A})v_1 = -(15\,\text{A})(60\,\text{V}) = -900$ W $= 900$ W(delivered)

[**c**] $p_{5\text{A}} = (5\,\text{A})v_2 = (5\,\text{A})(10\,\text{V}) = 50$ W$= -50$ W(delivered)

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AP 4.2 Redraw the circuit, choosing the node voltages and reference node as shown:



The two node voltage equations are:

$$-4.5 + \frac{v_1}{1} + \frac{v_1 - v_2}{6+2} = 0$$

$$\frac{v_2}{12} + \frac{v_2 - v_1}{6+2} + \frac{v_2 - 30}{4} = 0$$

Place these equations in standard form:

$$v_1\left(1 + \frac{1}{8}\right) + v_2\left(-\frac{1}{8}\right) = 4.5$$

$$v_1\left(-\frac{1}{8}\right) + v_2\left(\frac{1}{12} + \frac{1}{8} + \frac{1}{4}\right) = 7.5$$

Solving, $v_1 = 6$ V $\qquad v_2 = 18$ V

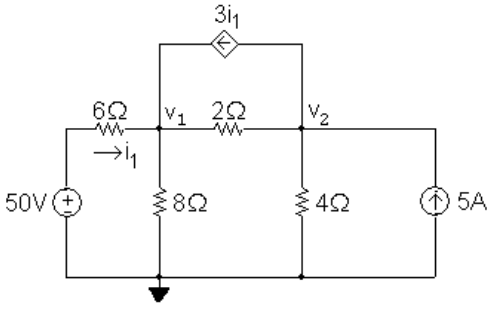
To find the voltage v, first find the current i through the series-connected $6\,\Omega$ and $2\,\Omega$ resistors:

$$i = \frac{v_1 - v_2}{6+2} = \frac{6 - 18}{8} = -1.5\,\text{A}$$

Using a KVL equation, calculate v:

$$v = 2i + v_2 = 2(-1.5) + 18 = 15\,\text{V}$$

AP 4.3 [a] Redraw the circuit, choosing the node voltages and reference node as shown:



The node voltage equations are:

$$\frac{v_1 - 50}{6} + \frac{v_1}{8} + \frac{v_1 - v_2}{2} - 3i_1 = 0$$

$$-5 + \frac{v_2}{4} + \frac{v_2 - v_1}{2} + 3i_1 = 0$$

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The dependent source requires the following constraint equation:

$$i_1 = \frac{50 - v_1}{6}$$

Place these equations in standard form:

$$\begin{array}{rcrcrcl} v_1\left(\frac{1}{6}+\frac{1}{8}+\frac{1}{2}\right) & + & v_2\left(-\frac{1}{2}\right) & + & i_1(-3) & = & \frac{50}{6} \\ v_1\left(-\frac{1}{2}\right) & + & v_2\left(\frac{1}{4}+\frac{1}{2}\right) & + & i_1(3) & = & 5 \\ v_1\left(\frac{1}{6}\right) & + & v_2(0) & + & i_1(1) & = & \frac{50}{6} \end{array}$$

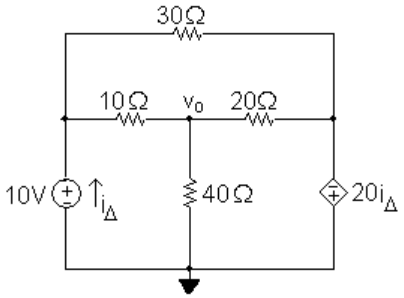
Solving, $v_1 = 32$ V; $v_2 = 16$ V; $i_1 = 3$ A

Using these values to calculate the power associated with each source:

$$\begin{aligned} p_{50\text{V}} = -50i_1 &= -150 \text{ W} \\ p_{5\text{A}} = -5(v_2) &= -80 \text{ W} \\ p_{3i_1} = 3i_1(v_2 - v_1) &= -144 \text{ W} \end{aligned}$$

[**b**] All three sources are delivering power to the circuit because the power computed in (a) for each of the sources is negative.

AP 4.4 Redraw the circuit and label the reference node and the node at which the node voltage equation will be written:



The node voltage equation is

$$\frac{v_o}{40} + \frac{v_o - 10}{10} + \frac{v_o + 20i_\Delta}{20} = 0$$

The constraint equation required by the dependent source is

$$i_\Delta = i_{10\Omega} + i_{30\Omega} = \frac{10 - v_o}{10} + \frac{10 + 20i_\Delta}{30}$$

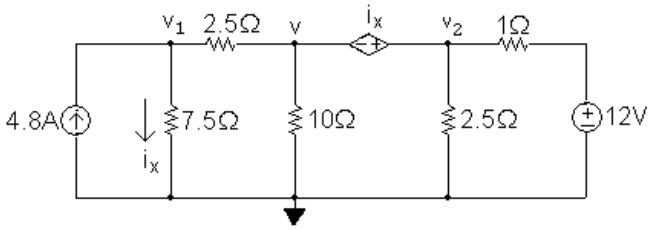
Place these equations in standard form:

$$\begin{array}{rcrcl} v_o\left(\frac{1}{40}+\frac{1}{10}+\frac{1}{20}\right) & + & i_\Delta(1) & = & 1 \\ v_o\left(\frac{1}{10}\right) & + & i_\Delta\left(1-\frac{20}{30}\right) & = & 1+\frac{10}{30} \end{array}$$

Solving, $i_\Delta = -3.2$ A and $v_o = 24$ V

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AP 4.5 Redraw the circuit identifying the three node voltages and the reference node:



Note that the dependent voltage source and the node voltages v and v_2 form a supernode. The v_1 node voltage equation is

$$\frac{v_1}{7.5} + \frac{v_1 - v}{2.5} - 4.8 = 0$$

The supernode equation is

$$\frac{v - v_1}{2.5} + \frac{v}{10} + \frac{v_2}{2.5} + \frac{v_2 - 12}{1} = 0$$

The constraint equation due to the dependent source is

$$i_x = \frac{v_1}{7.5}$$

The constraint equation due to the supernode is

$$v + i_x = v_2$$

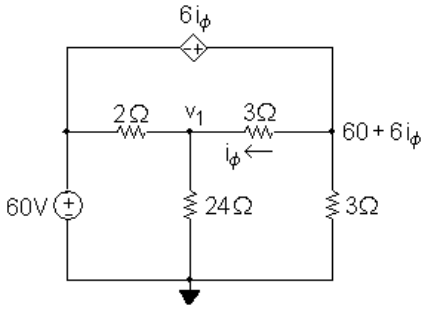
Place this set of equations in standard form:

$$\begin{array}{lcllcllcl}
v_1\left(\frac{1}{7.5}+\frac{1}{2.5}\right) & + & v\left(-\frac{1}{2.5}\right) & + & v_2(0) & + & i_x(0) & = & 4.8 \\
v_1\left(-\frac{1}{2.5}\right) & + & v\left(\frac{1}{2.5}+\frac{1}{10}\right) & + & v_2\left(\frac{1}{2.5}+1\right) & + & i_x(0) & = & 12 \\
v_1\left(-\frac{1}{7.5}\right) & + & v(0) & + & v_2(0) & + & i_x(1) & = & 0 \\
v_1(0) & + & v(1) & + & v_2(-1) & + & i_x(1) & = & 0
\end{array}$$

Solving this set of equations gives $v_1 = 15$ V, $v_2 = 10$ V, $i_x = 2$ A, and $v = 8$ V.

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AP 4.6 Redraw the circuit identifying the reference node and the two unknown node voltages. Note that the right-most node voltage is the sum of the 60 V source and the dependent source voltage.



The node voltage equation at v_1 is

$$\frac{v_1 - 60}{2} + \frac{v_1}{24} + \frac{v_1 - (60 + 6i_\phi)}{3} = 0$$

The constraint equation due to the dependent source is

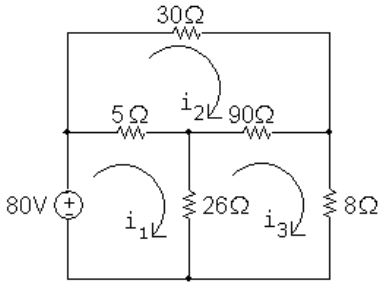
$$i_\phi = \frac{60 + 6i_\phi - v_1}{3}$$

Place these two equations in standard form:

$$\begin{aligned} v_1\left(\frac{1}{2} + \frac{1}{24} + \frac{1}{3}\right) \quad &+ \quad i_\phi(-2) \quad &= \quad 30 + 20 \\ v_1\left(\frac{1}{3}\right) \quad &+ \quad i_\phi(1-2) \quad &= \quad 20 \end{aligned}$$

Solving, $\quad i_\phi = -4$ A $\quad$ and $\quad v_1 = 48$ V

AP 4.7 [**a**] Redraw the circuit identifying the three mesh currents:



The mesh current equations are:

$$\begin{aligned} -80 + 5(i_1 - i_2) + 26(i_1 - i_3) \quad &= \quad 0 \\ 30i_2 + 90(i_2 - i_3) + 5(i_2 - i_1) \quad &= \quad 0 \\ 8i_3 + 26(i_3 - i_1) + 90(i_3 - i_2) \quad &= \quad 0 \end{aligned}$$

Place these equations in standard form:

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$$\begin{aligned} 31i_1 - 5i_2 - 26i_3 &= 80 \\ -5i_1 + 125i_2 - 90i_3 &= 0 \\ -26i_1 - 90i_2 + 124i_3 &= 0 \end{aligned}$$

Solving,

$i_1 = 5\,\text{A}; \qquad i_2 = 2\,\text{A}; \qquad i_3 = 2.5\,\text{A}$

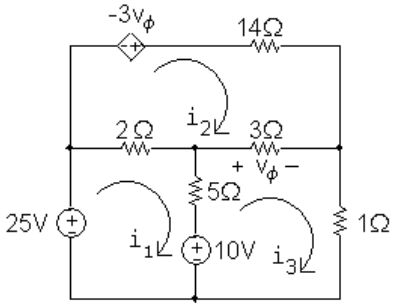
$p_{80V} = -(80)i_1 = -(80)(5) = -400\,\text{W}$

Therefore the 80 V source is delivering 400 W to the circuit.

[b] $p_{8\Omega} = (8)i_3^2 = 8(2.5)^2 = 50$ W, so the $8\,\Omega$ resistor dissipates 50 W.

AP 4.8 **[a]** $b = 8, \qquad n = 6, \qquad b - n + 1 = 3$

[b] Redraw the circuit identifying the three mesh currents:



The three mesh-current equations are

$$\begin{aligned} -25 + 2(i_1 - i_2) + 5(i_1 - i_3) + 10 &= 0 \\ -(-3v_\phi) + 14i_2 + 3(i_2 - i_3) + 2(i_2 - i_1) &= 0 \\ 1i_3 - 10 + 5(i_3 - i_1) + 3(i_3 - i_2) &= 0 \end{aligned}$$

The dependent source constraint equation is

$v_\phi = 3(i_3 - i_2)$

Place these four equations in standard form:

$$\begin{aligned} 7i_1 - 2i_2 - 5i_3 + 0v_\phi &= 15 \\ -2i_1 + 19i_2 - 3i_3 + 3v_\phi &= 0 \\ -5i_1 - 3i_2 + 9i_3 + 0v_\phi &= 10 \\ 0i_1 + 3i_2 - 3i_3 + 1v_\phi &= 0 \end{aligned}$$

Solving

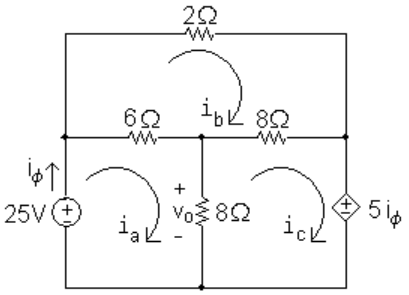
$i_1 = 4\,\text{A}; \qquad i_2 = -1\,\text{A}; \qquad i_3 = 3\,\text{A}; \qquad v_\phi = 12\,\text{V}$

$p_{\text{ds}} = -(-3v_\phi)i_2 = 3(12)(-1) = -36\,\text{W}$

Thus, the dependent source is delivering 36 W, or absorbing −36 W.

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AP 4.9 Redraw the circuit identifying the three mesh currents:



The mesh current equations are:

$$\begin{aligned} -25 + 6(i_\mathrm{a} - i_\mathrm{b}) + 8(i_\mathrm{a} - i_\mathrm{c}) &= 0 \\ 2i_\mathrm{b} + 8(i_\mathrm{b} - i_\mathrm{c}) + 6(i_\mathrm{b} - i_\mathrm{a}) &= 0 \\ 5i_\phi + 8(i_\mathrm{c} - i_\mathrm{a}) + 8(i_\mathrm{c} - i_\mathrm{b}) &= 0 \end{aligned}$$

The dependent source constraint equation is $i_\phi = i_\mathrm{a}$. We can substitute this simple expression for i_ϕ into the third mesh equation and place the equations in standard form:

$$\begin{aligned} 14i_\mathrm{a} - 6i_\mathrm{b} - 8i_\mathrm{c} &= 25 \\ -6i_\mathrm{a} + 16i_\mathrm{b} - 8i_\mathrm{c} &= 0 \\ -3i_\mathrm{a} - 8i_\mathrm{b} + 16i_\mathrm{c} &= 0 \end{aligned}$$

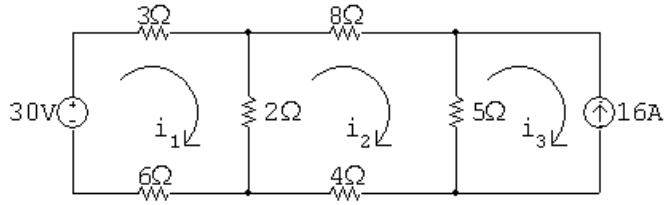
Solving,

$$i_\mathrm{a} = 4\,\mathrm{A}; \qquad i_\mathrm{b} = 2.5\,\mathrm{A}; \qquad i_\mathrm{c} = 2\,\mathrm{A}$$

Thus,

$$v_o = 8(i_\mathrm{a} - i_\mathrm{c}) = 8(4 - 2) = 16\,\mathrm{V}$$

AP 4.10 Redraw the circuit identifying the mesh currents:



Since there is a current source on the perimeter of the i_3 mesh, we know that $i_3 = -16$ A. The remaining two mesh equations are

$$\begin{aligned} -30 + 3i_1 + 2(i_1 - i_2) + 6i_1 &= 0 \\ 8i_2 + 5(i_2 + 16) + 4i_2 + 2(i_2 - i_1) &= 0 \end{aligned}$$

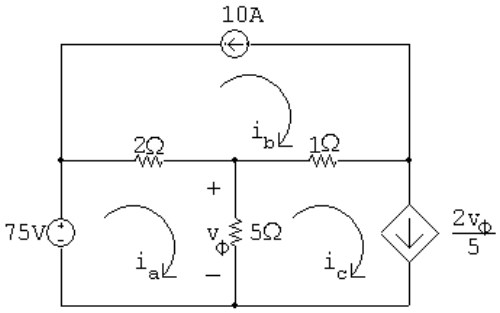
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Place these equations in standard form:

$$\begin{aligned} 11i_1 - 2i_2 &= 30 \\ -2i_1 + 19i_2 &= -80 \end{aligned}$$

Solving: $i_1 = 2\,\text{A}, \quad i_2 = -4\,\text{A}, \quad i_3 = -16\,\text{A}$
The current in the $2\,\Omega$ resistor is $i_1 - i_2 = 6\,\text{A}$ $\therefore$ $p_{2\,\Omega} = (6)^2(2) = 72\,\text{W}$
Thus, the $2\,\Omega$ resistors dissipates 72 W.

AP 4.11 Redraw the circuit and identify the mesh currents:



There are current sources on the perimeters of both the i_b mesh and the i_c mesh, so we know that

$$i_\text{b} = -10\,\text{A}; \qquad i_\text{c} = \frac{2v_\phi}{5}$$

The remaining mesh current equation is

$$-75 + 2(i_\text{a} + 10) + 5(i_\text{a} - 0.4v_\phi) = 0$$

The dependent source requires the following constraint equation:

$$v_\phi = 5(i_\text{a} - i_\text{c}) = 5(i_\text{a} - 0.4v_\phi)$$

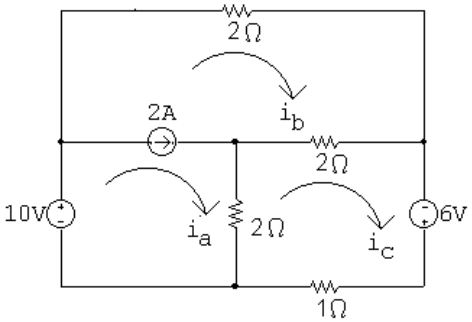
Place the mesh current equation and the dependent source equation is standard form:

$$\begin{aligned} 7i_\text{a} - 2v_\phi &= 55 \\ 5i_\text{a} - 3v_\phi &= 0 \end{aligned}$$

Solving: $i_\text{a} = 15\,\text{A}; \qquad i_\text{b} = -10\,\text{A}; \qquad i_\text{c} = 10\,\text{A}; \qquad v_\phi = 25\,\text{V}$
Thus, $i_\text{a} = 15$ A.

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AP 4.12 Redraw the circuit and identify the mesh currents:



The 2 A current source is shared by the meshes i_a and i_b. Thus we combine these meshes to form a supermesh and write the following equation:

$$-10 + 2i_\mathrm{b} + 2(i_\mathrm{b} - i_\mathrm{c}) + 2(i_\mathrm{a} - i_\mathrm{c}) = 0$$

The other mesh current equation is

$$-6 + 1i_\mathrm{c} + 2(i_\mathrm{c} - i_\mathrm{a}) + 2(i_\mathrm{c} - i_\mathrm{b}) = 0$$

The supermesh constraint equation is

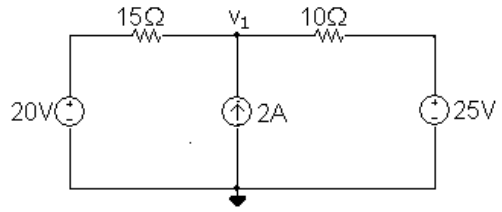
$$i_\mathrm{a} - i_\mathrm{b} = 2$$

Place these three equations in standard form:

$$\begin{aligned} 2i_\mathrm{a} + 4i_\mathrm{b} - 4i_\mathrm{c} &= 10 \\ -2i_\mathrm{a} - 2i_\mathrm{b} + 5i_\mathrm{c} &= 6 \\ i_\mathrm{a} - i_\mathrm{b} + 0i_\mathrm{c} &= 2 \end{aligned}$$

Solving, $i_\mathrm{a} = 7\,\mathrm{A}$; $i_\mathrm{b} = 5\,\mathrm{A}$; $i_\mathrm{c} = 6\,\mathrm{A}$
Thus, $p_{1\,\Omega} = i_\mathrm{c}^2(1) = (6)^2(1) = 36\,\mathrm{W}$

AP 4.13 Redraw the circuit and identify the reference node and the node voltage v_1:



The node voltage equation is

$$\frac{v_1 - 20}{15} - 2 + \frac{v_1 - 25}{10} = 0$$

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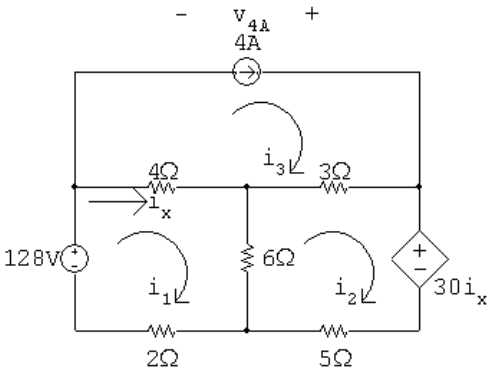
Rearranging and solving,

$$v_1\left(\frac{1}{15}+\frac{1}{10}\right)=2+\frac{20}{15}+\frac{25}{10} \qquad \therefore \quad v_1=35\,\text{V}$$

$$p_{2A}=-35(2)=-70\,\text{W}$$

Thus the 2 A current source delivers 70 W.

AP 4.14 Redraw the circuit and identify the mesh currents:



There is a current source on the perimeter of the i_3 mesh, so $i_3=4$ A. The other two mesh current equations are

$$\begin{aligned} -128+4(i_1-4)+6(i_1-i_2)+2i_1 &= 0 \\ 30i_x+5i_2+6(i_2-i_1)+3(i_2-4) &= 0 \end{aligned}$$

The constraint equation due to the dependent source is

$$i_x=i_1-i_3=i_1-4$$

Substitute the constraint equation into the second mesh equation and place the resulting two mesh equations in standard form:

$$\begin{aligned} 12i_1-6i_2 &= 144 \\ 24i_1+14i_2 &= 132 \end{aligned}$$

Solving,

$$i_1=9\,\text{A}; \qquad i_2=-6\,\text{A}; \qquad i_3=4\,\text{A}; \qquad i_x=9-4=5\,\text{A}$$

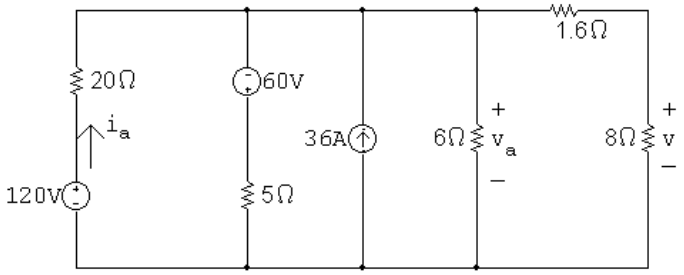
$$\therefore \quad v_{4\text{A}}=3(i_3-i_2)-4i_x=10\,\text{V}$$

$$p_{4A}=-v_{4\text{A}}(4)=-(10)(4)=-40\,\text{W}$$

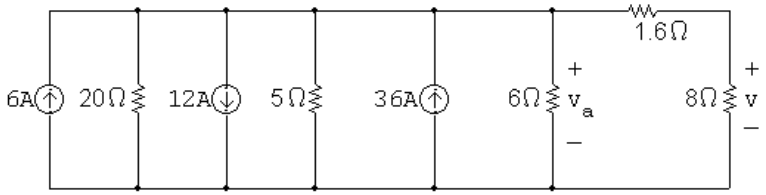
Thus, the 2 A current source delivers 40 W.

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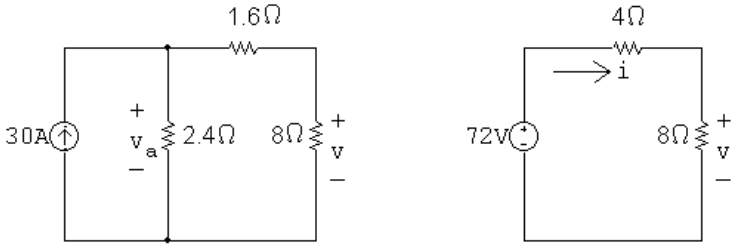
AP 4.15 [**a**] Redraw the circuit with a helpful voltage and current labeled:



Transform the 120 V source in series with the 20 Ω resistor into a 6 A source in parallel with the 20 Ω resistor. Also transform the −60 V source in series with the 5 Ω resistor into a −12 A source in parallel with the 5 Ω resistor. The result is the following circuit:



Combine the three current sources into a single current source, using KCL, and combine the 20 Ω, 5 Ω, and 6 Ω resistors in parallel. The resulting circuit is shown on the left. To simplify the circuit further, transform the resulting 30 A source in parallel with the 2.4 Ω resistor into a 72 V source in series with the 2.4 Ω resistor. Combine the 2.4 Ω resistor in series with the 1.6 Ω resisor to get a very simple circuit that still maintains the voltage v. The resulting circuit is on the right.



Use voltage division in the circuit on the right to calculate v as follows:

$$v = \frac{8}{12}(72) = 48\,\text{V}$$

[**b**] Calculate i in the circuit on the right using Ohm's law:

$$i = \frac{v}{8} = \frac{48}{8} = 6\,\text{A}$$

Now use i to calculate v_a in the circuit on the left:

$$v_\text{a} = 6(1.6 + 8) = 57.6\,\text{V}$$

Returning back to the original circuit, note that the voltage v_a is also the voltage drop across the series combination of the 120 V source and 20 Ω

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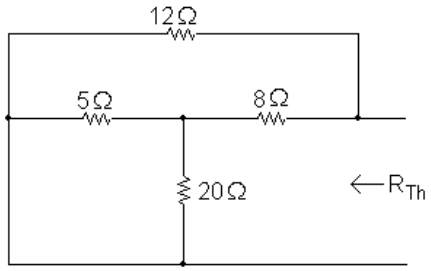
resistor. Use this fact to calculate the current in the 120 V source, i_a:

$$i_a = \frac{120 - v_a}{20} = \frac{120 - 57.6}{20} = 3.12\,\text{A}$$

$$p_{120V} = -(120)i_a = -(120)(3.12) = -374.40\,\text{W}$$

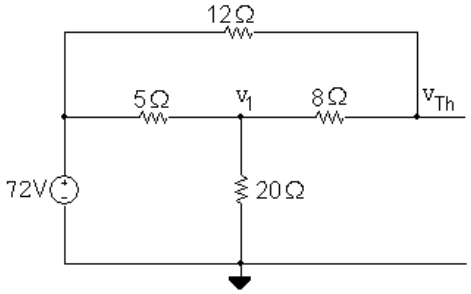
Thus, the 120 V source delivers 374.4 W.

AP 4.16 To find R_{Th}, replace the 72 V source with a short circuit:



Note that the $5\,\Omega$ and $20\,\Omega$ resistors are in parallel, with an equivalent resistance of $5\|20 = 4\,\Omega$. The equivalent $4\,\Omega$ resistance is in series with the $8\,\Omega$ resistor for an equivalent resistance of $4 + 8 = 12\,\Omega$. Finally, the $12\,\Omega$ equivalent resistance is in parallel with the $12\,\Omega$ resistor, so $R_{\text{Th}} = 12\|12 = 6\,\Omega$.

Use node voltage analysis to find v_{Th}. Begin by redrawing the circuit and labeling the node voltages:



The node voltage equations are

$$\frac{v_1 - 72}{5} + \frac{v_1}{20} + \frac{v_1 - v_{\text{Th}}}{8} = 0$$

$$\frac{v_{\text{Th}} - v_1}{8} + \frac{v_{\text{Th}} - 72}{12} = 0$$

Place these equations in standard form:

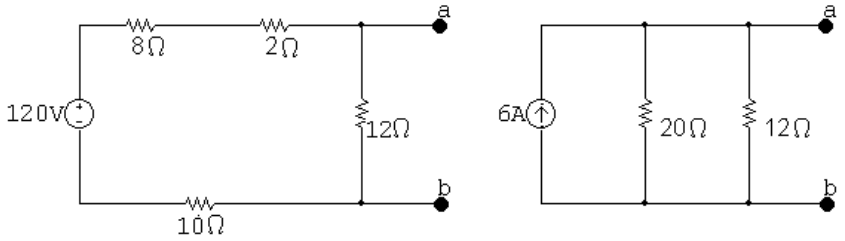
$$v_1\left(\frac{1}{5} + \frac{1}{20} + \frac{1}{8}\right) + v_{\text{Th}}\left(-\frac{1}{8}\right) = \frac{72}{5}$$

$$v_1\left(-\frac{1}{8}\right) + v_{\text{Th}}\left(\frac{1}{8} + \frac{1}{12}\right) = 6$$

Solving, $v_1 = 60$ V and $v_{\text{Th}} = 64.8$ V. Therefore, the Thévenin equivalent circuit is a 64.8 V source in series with a $6\,\Omega$ resistor.

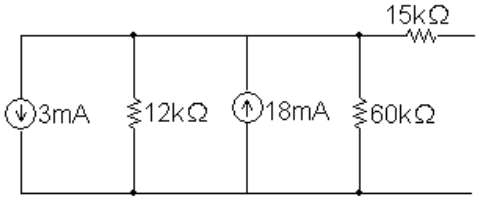
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AP 4.17 We begin by performing a source transformation, turning the parallel combination of the 15 A source and 8 Ω resistor into a series combination of a 120 V source and an 8 Ω resistor, as shown in the figure on the left. Next, combine the 2 Ω, 8 Ω and 10 Ω resistors in series to give an equivalent 20 Ω resistance. Then transform the series combination of the 120 V source and the 20 Ω equivalent resistance into a parallel combination of a 6 A source and a 20 Ω resistor, as shown in the figure on the right.

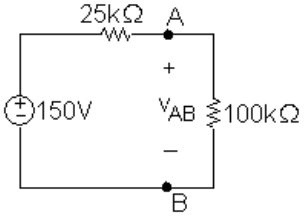


Finally, combine the 20 Ω and 12 Ω parallel resistors to give $R_N = 20\|12 = 7.5\,\Omega$. Thus, the Norton equivalent circuit is the parallel combination of a 6 A source and a 7.5 Ω resistor.

AP 4.18 Find the Thévenin equivalent with respect to A, B using source transformations. To begin, convert the series combination of the −36 V source and 12 kΩ resistor into a parallel combination of a −3 mA source and 12 kΩ resistor. The resulting circuit is shown below:



Now combine the two parallel current sources and the two parallel resistors to give a $-3 + 18 = 15$ mA source in parallel with a 12 k$\|$60 k= 10 kΩ resistor. Then transform the 15 mA source in parallel with the 10 kΩ resistor into a 150 V source in series with a 10 kΩ resistor, and combine this 10 kΩ resistor in series with the 15 kΩ resistor. The Thévenin equivalent is thus a 150 V source in series with a 25 kΩ resistor, as seen to the left of the terminals A,B in the circuit below.

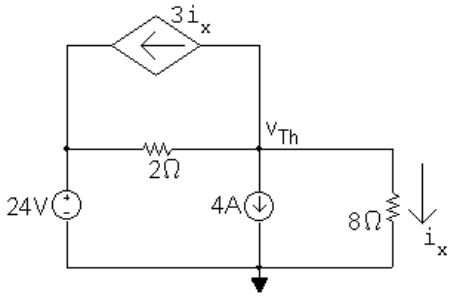


Now attach the voltmeter, modeled as a 100 kΩ resistor, to the Thévenin equivalent and use voltage division to calculate the meter reading v_{AB}:

$$v_{AB} = \frac{100{,}000}{125{,}000}(150) = 120\,\text{V}$$

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AP 4.19 Begin by calculating the open circuit voltage, which is also v_{Th}, from the circuit below:



Summing the currents away from the node labeled v_{Th} We have

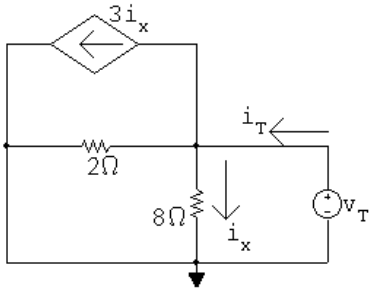
$$\frac{v_{\text{Th}}}{8} + 4 + 3i_x + \frac{v_{\text{Th}} - 24}{2} = 0$$

Also, using Ohm's law for the $8\,\Omega$ resistor,

$$i_x = \frac{v_{\text{Th}}}{8}$$

Substituting the second equation into the first and solving for v_{Th} yields $v_{\text{Th}} = 8\,\text{V}$.

Now calculate R_{Th}. To do this, we use the test source method. Replace the voltage source with a short circuit, the current source with an open circuit, and apply the test voltage v_{T}, as shown in the circuit below:



Write a KCL equation at the middle node:

$$i_{\text{T}} = i_x + 3i_x + v_{\text{T}}/2 = 4i_x + v_{\text{T}}/2$$

Use Ohm's law to determine i_x as a function of v_{T}:

$$i_x = v_{\text{T}}/8$$

Substitute the second equation into the first equation:

$$i_{\text{T}} = 4(v_{\text{T}}/8) + v_{\text{T}}/2 = v_{\text{T}}$$

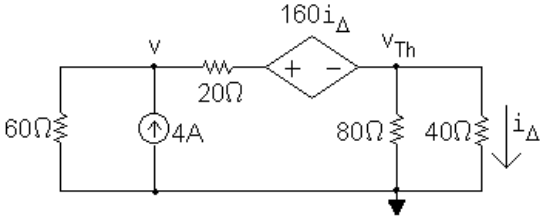
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Thus,

$$R_{\text{Th}} = v_{\text{T}}/i_{\text{T}} = 1\,\Omega$$

The Thévenin equivalent is an 8 V source in series with a 1 Ω resistor.

AP 4.20 Begin by calculating the open circuit voltage, which is also v_{Th}, using the node voltage method in the circuit below:



The node voltage equations are

$$\begin{aligned} \frac{v}{60} + \frac{v - (v_{\text{Th}} + 160 i_\Delta)}{20} - 4 &= 0, \\ \frac{v_{\text{Th}}}{40} + \frac{v_{\text{Th}}}{80} + \frac{v_{\text{Th}} + 160 i_\Delta - v}{20} &= 0 \end{aligned}$$

The dependent source constraint equation is

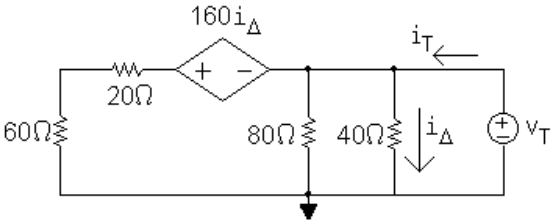
$$i_\Delta = \frac{v_{\text{Th}}}{40}$$

Substitute the constraint equation into the node voltage equations and put the two equations in standard form:

$$\begin{aligned} v\left(\frac{1}{60} + \frac{1}{20}\right) + v_{\text{Th}}\left(-\frac{5}{20}\right) &= 4 \\ v\left(-\frac{1}{20}\right) + v_{\text{Th}}\left(\frac{1}{40} + \frac{1}{80} + \frac{5}{20}\right) &= 0 \end{aligned}$$

Solving, $v = 172.5$ V and $v_{\text{Th}} = 30$ V.

Now use the test source method to calculate the test current and thus R_{Th}. Replace the current source with a short circuit and apply the test source to get the following circuit:



Write a KCL equation at the rightmost node:

$$i_{\text{T}} = \frac{v_{\text{T}}}{80} + \frac{v_{\text{T}}}{40} + \frac{v_{\text{T}} + 160 i_\Delta}{80}$$

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The dependent source constraint equation is

$$i_\Delta = \frac{v_\mathrm{T}}{40}$$

Substitute the constraint equation into the KCL equation and simplify the right-hand side:

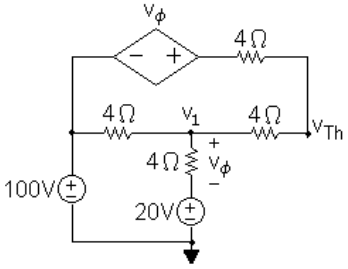
$$i_\mathrm{T} = \frac{v_\mathrm{T}}{10}$$

Therefore,

$$R_\mathrm{Th} = \frac{v_\mathrm{T}}{i_\mathrm{T}} = 10\,\Omega$$

Thus, the Thévenin equivalent is a 30 V source in series with a $10\,\Omega$ resistor.

AP 4.21 First find the Thévenin equivalent circuit. To find v_Th, create an open circuit between nodes a and b and use the node voltage method with the circuit below:



The node voltage equations are:

$$\begin{aligned} \frac{v_\mathrm{Th} - (100 + v_\phi)}{4} + \frac{v_\mathrm{Th} - v_1}{4} &= 0 \\ \frac{v_1 - 100}{4} + \frac{v_1 - 20}{4} + \frac{v_1 - v_\mathrm{Th}}{4} &= 0 \end{aligned}$$

The dependent source constraint equation is

$$v_\phi = v_1 - 20$$

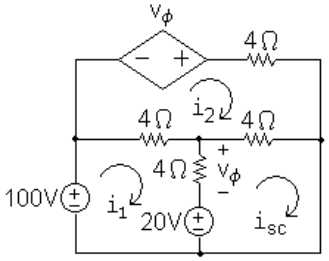
Place these three equations in standard form:

$$\begin{aligned} v_\mathrm{Th}\left(\frac{1}{4} + \frac{1}{4}\right) + v_1\left(-\frac{1}{4}\right) + v_\phi\left(-\frac{1}{4}\right) &= 25 \\ v_\mathrm{Th}\left(-\frac{1}{4}\right) + v_1\left(\frac{1}{4} + \frac{1}{4} + \frac{1}{4}\right) + v_\phi\,(0) &= 30 \\ v_\mathrm{Th}\,(0) + v_1\,(1) + v_\phi\,(-1) &= 20 \end{aligned}$$

Solving, $v_\mathrm{Th} = 120$ V, $v_1 = 80$ V, and $v_\phi = 60$ V.

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Now create a short circuit between nodes a and b and use the mesh current method with the circuit below:



The mesh current equations are

$$\begin{aligned} -100 + 4(i_1 - i_2) + v_\phi + 20 &= 0 \\ -v_\phi + 4i_2 + 4(i_2 - i_{\text{sc}}) + 4(i_2 - i_1) &= 0 \\ -20 - v_\phi + 4(i_{\text{sc}} - i_2) &= 0 \end{aligned}$$

The dependent source constraint equation is

$$v_\phi = 4(i_1 - i_{\text{sc}})$$

Place these four equations in standard form:

$$\begin{aligned} 4i_1 - 4i_2 + 0i_{\text{sc}} + v_\phi &= 80 \\ -4i_1 + 12i_2 - 4i_{\text{sc}} - v_\phi &= 0 \\ 0i_1 - 4i_2 + 4i_{\text{sc}} - v_\phi &= 20 \\ 4i_1 + 0i_2 - 4i_{\text{sc}} - v_\phi &= 0 \end{aligned}$$

Solving, $i_1 = 45$ A, $i_2 = 30$ A, $i_{\text{sc}} = 40$ A, and $v_\phi = 20$ V. Thus,

$$R_{\text{Th}} = \frac{v_{\text{Th}}}{i_{\text{sc}}} = \frac{120}{40} = 3\,\Omega$$

[**a**] For maximum power transfer, $R = R_{\text{Th}} = 3\,\Omega$

[**b**] The Thévenin voltage, $v_{\text{Th}} = 120$ V, splits equally between the Thévenin resistance and the load resistance, so

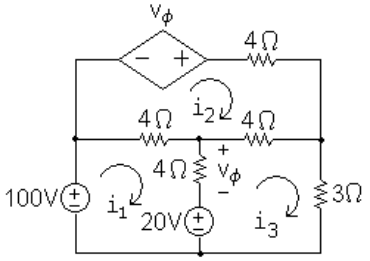
$$v_{\text{load}} = \frac{120}{2} = 60\,\text{V}$$

Therefore,

$$p_{\max} = \frac{v_{\text{load}}^2}{R_{\text{load}}} = \frac{60^2}{3} = 1200\,\text{W}$$

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AP 4.22 Sustituting the value $R = 3\,\Omega$ into the circuit and identifying three mesh currents we have the circuit below:



The mesh current equations are:

$$\begin{aligned} -100 + 4(i_1 - i_2) + v_\phi + 20 &= 0 \\ -v_\phi + 4i_2 + 4(i_2 - i_3) + 4(i_2 - i_1) &= 0 \\ -20 - v_\phi + 4(i_3 - i_2) + 3i_3 &= 0 \end{aligned}$$

The dependent source constraint equation is

$$v_\phi = 4(i_1 - i_3)$$

Place these four equations in standard form:

$$\begin{aligned} 4i_1 - 4i_2 + 0i_3 + v_\phi &= 80 \\ -4i_1 + 12i_2 - 4i_3 - v_\phi &= 0 \\ 0i_1 - 4i_2 + 7i_3 - v_\phi &= 20 \\ 4i_1 + 0i_2 - 4i_3 - v_\phi &= 0 \end{aligned}$$

Solving, $i_1 = 30$ A, $i_2 = 20$ A, $i_3 = 20$ A, and $v_\phi = 40$ V.

[a] $p_{100\text{V}} = -(100)i_1 = -(100)(30) = -3000$ W. Thus, the 100 V source is delivering 3000 W.

[b] $p_{\text{depsource}} = -v_\phi i_2 = -(40)(20) = -800$ W. Thus, the dependent source is delivering 800 W.

[c] From Assessment Problem 4.21(b), the power delivered to the load resistor is 1200 W, so the load power is (1200/3800)100 = 31.58% of the combined power generated by the 100 V source and the dependent source.

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